

sabinaHSBM Docker Setup

Overview

This guide will help Windows users run the *sabinaHSBM* package within a Linux-based Docker container. While this tutorial is tailored for Windows, similar instructions should be followed for Mac or Linux users.

What's included in the Docker Image

- **Linux operating system:** Ubuntu 24.04
- **R Environment:** Pre-configured with all necessary packages, including *reticulate* for Python integration.
- **Python & graph-tool:** Python is pre-configured with *graph-tool* for network analysis.
- **sabinaHSBM:** The Docker image includes the *sabinaHSBM* package and all required dependencies.
- **RStudio Server:** Installed and ready to use as a browser-based R interface (user creation recommended).
- **Jupyter Notebook:** Pre-installed and ready to launch from the container for interactive coding and analysis.

Step-by-step setup

1. Install Docker Desktop

Download and install Docker Desktop for Windows from <https://www.docker.com/products/docker-desktop>. Registration and sign-in are required. Once Docker Desktop is installed, you can open it and run the following commands either from its integrated terminal (> **Terminal** button) or from the Windows Command Prompt (CMD) or PowerShell.

2. Pull the docker image

To download the pre-configured image from Docker Hub, run:

```
docker pull herlima/sabinahsbm
```

3. Create and start the container

The following command creates and runs a new Docker container named *sabinahsbm_container* using the *sabinahsbm* image. The *-p* flags are optional and map ports 8787, 8880 from the container to the host machine, enabling access to Jupyter Notebook or RStudio Server, respectively, if needed. The *-v* flag is also optional (but recommended) and establishes a bind mount between your local directory (*/local/path/to/your/project*) and the container's bind-mounted directory (*/home/my_project*), allowing seamless file synchronization. Files created or modified in the bind-mounted directory will be directly accessible

on your local machine. Alternatively, you can use a Docker volume instead of a bind mount. Unlike bind mounts, which link directly to local directory, volume are stored within a Docker filesystem.

```
docker run -it --name sabinahsbm_container \
  -p 8787:8787 -p 8880:8880 \
  -v "local/path/to/your/project:/home/my_project" \
  sabinahsbm bash
```

After executing this command, you will enter an interactive shell, providing direct access to the container's command line.

4. Run R, RStudio Server or Jupyter Notebook

Once the docker container is running, you can interact with the *sabinaHSBM* package using different environments: through the R console, RStudio Server, or Jupyter Notebook.

- **Run R from the terminal**

Start an R session by typing `R`. This opens an interactive console where you can load and use *sabinaHSBM* from the terminal environment.

- **Use RStudio Server** (*optional*)

RStudio Server is already installed in the image.

A default user was created with username `test` and password `sabinahsbm`. But we recommend you to set your own. Create a user and password (This step is only needed the first time) (replace `yourname` with your desired username and password):

```
useradd -m yourname
passwd yourname # you will be prompted to set a password
```

Then launch RStudio Server:

```
/usr/lib/rstudio-server/bin/rsserver --server-daemonize=0 \
  --www-port=8787 --www-address=0.0.0.0
```

In your browser, open: `http://localhost:8787`

Log in with the user and password you just created, or use the default credentials.

To work in your mounted project directory `/home/my_project` use in the R console `setwd("/home/my_project");` or go to the Files pane (bottom-right), click `"..."` → `"Go to folder..."` and enter `/home/my_project`.

Note: RStudio Server runs in the foreground. Keep the terminal open while working.

- **Use Jupyter Notebook** (*optional*)

You can also use a Jupyter Notebook. Launch it with:

```
jupyter notebook --allow-root --no-browser \  
--ip 0.0.0.0 --port=8880
```

The terminal will display a clickable URL. Simply click the provided link or paste it into your browser to access the Jupyter interface, ideal for coding, data visualization, and interactive analysis.

5. Use the *sabinaHSBM* package

Here's an example to get started:

```
setwd("/home/my_project")  
  
# Load sabinaHSBM  
library(sabinaHSBM)  
  
# Load the data  
data(dat, package = "sabinaHSBM")  
  
# Prepare the input  
myInput <- hsbm.input(  
  dat,  
  n_folds = 10  
)  
  
# Generate link predictions  
myPred <- hsbm.predict(myInput,  
  iter = 1000,  
  method = "conditional_missing"  
)  
  
# Reconstruct the network and evaluate  
myReconst <- hsbm.reconstructed(myPred,  
  rm_documented = TRUE,  
  threshold = "prc_closest_topright")  
  
summary(myReconst)  
  
# Exit R  
q()
```

6. Exit the Container

To exit the container, simply type:

```
exit
```

7. Stop and Restart the container

To stop the container:

```
docker stop sabinahsbm_container
```

To start the container again and access its shell, run:

```
docker start -ai sabinahsbm_container  
docker exec -it sabinahsbm_container bash
```